

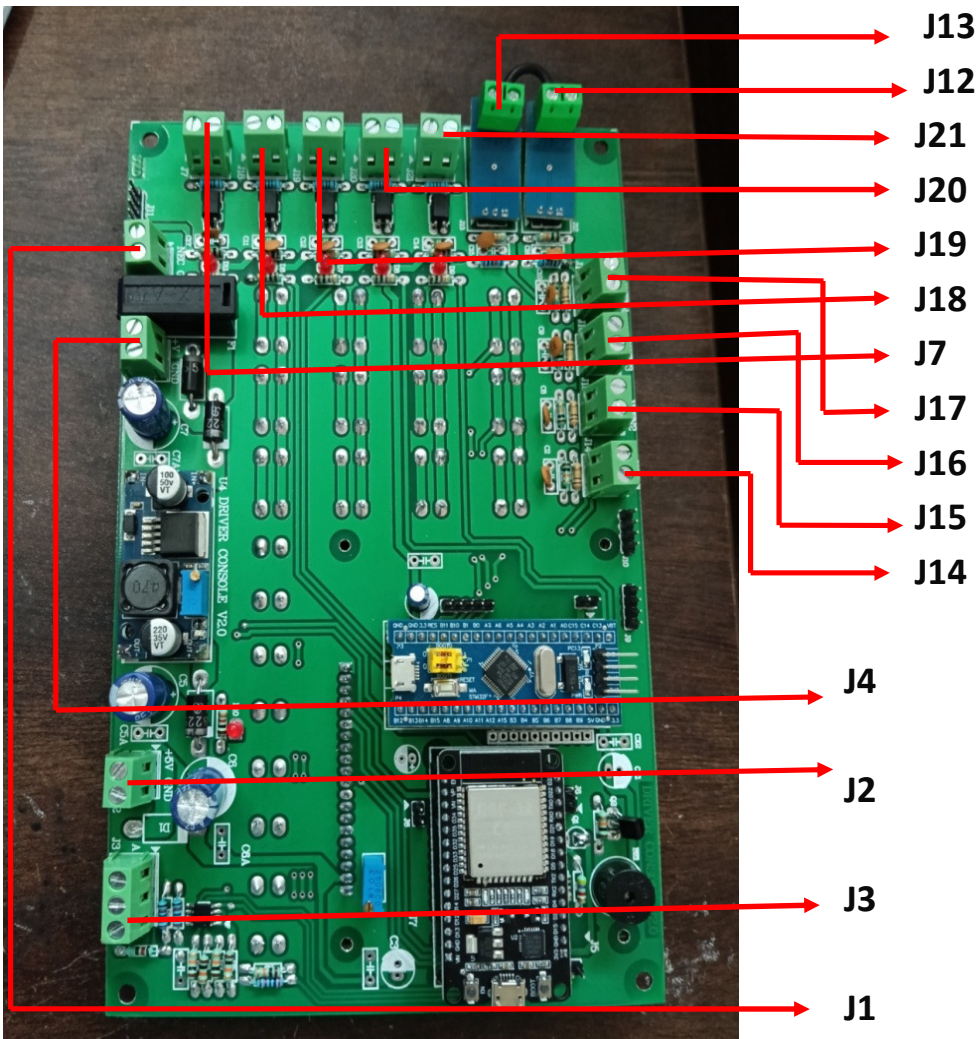
Bus Driver Console V2.0 – Wiring Manual

Document Version: 1.0

1. Overview

The **Bus Driver Console V2.0** is a dual-microcontroller driver board (STM32 & ESP32) designed for flexibility. This document serves as a wiring guide for all terminal blocks and connection points.

2. Terminal Block CONNECTORS



3. Terminal Label Mapping & Function

Note: Ensure power is off before wiring.

Power Section (Left side)

- **J2:** Regulated 5V Input for the Board.
- **J4 – (+24V GND):** Main power Supply IN.

Logic Inputs (Top Row)

Health Packet Inputs and Tamper Input

- **J7:** Opto Coupler Channel 1 (Tamper)
- **J18:** Opto Coupler Channel 2
- **J19:** Opto Coupler Channel 3
- **J20:** Opto Coupler Channel 4
- **J21:** Opto Coupler Channel 5

These are the **5 stacked terminals on top**

Current Sensors (Top-Right)

- **J12 & J13:** Current sensors for LED board monitor.

Sensor/ Signal Inputs (Bottom Row)

Pin out: VCC (3.3V/5V) | GND | Signal

- **J14, J15, J16, J17:** 2-pin terminals (GND, Signal) for Voltage reading (Camera).

Communication

- **J3 – (A – GND - B):** RS485 Communication to LED boards.
 - **J1:** NEC IR wire output for LED boards
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4. Summary Table

Label	Function	Type
J4	+24V Main Power	Power
J2	+5V Aux Power	Power
J1	NEC IR	Output Digital
J7, J18-J21	Channels 1-5	Input Digital
J12-J13	Current Sensor	Input Analog
J14-J17	Sensor Inputs	Input Analog
J3	RS485	Output Digital

5. Safety & Installation Notes

- **High Voltage Warning:** Disconnect the +24V supply (J4) before touching load outputs to avoid arching or damage.
- **Polarity Check:** Reversing polarity on J4 or J2 will cause permanent board failure. Double-check + and - markings.
- **Wire Gauge:** Use minimum 14-16 AWG wire for J4, J2, J12 and J13 to handle high current loads.
- **Status Indicators:** Red LEDs (D6-D9, D11) indicate channel activation.

6. Assigned Terminal Connectors

J14 --- Camera.

J13, J12 --- LED display 1, 2.

J7 --- Tamper.